Networksecurity 2

Rubric 4

Network Access & Authentication
The project must demonstrate basic Network Access Control implementation in a lab environment. Identity-based access must be configured using username/password or certificate authentication. A basic site-to-site VPN must be configured between two virtual networks. Role-based access control must be implemented for at least two different user types. All configurations must include screenshots and basic setup documentation.

1. VirtualBox Network Architecture

To isolate traffic properly and simulate a true Site-to-Site environment on a single physical host, configure your VirtualBox network adapters using **Internal Networks**.

Virtual Machine	Adapter 1 (WAN / Management)	Adapter 2 (HQ LAN)	Adapter 3 (Branch LAN)
OPNsense Firewall	**Bridged** or **NAT** (Internet Access)	**Internal Network**: `hq_lan`	**Internal Network**: `branch_lan`
Ubuntu Client (HQ Staff)	**Internal Network**: `hq_lan`	None	None
Kali Linux	**Internal Network**: `hq_lan`	None	None

(HQ Admin)

Note: In OPNsense, we will assign the `branch_lan` interface to act as a simulated remote branch office so you can build a loopback Site-to-Site VPN directly inside the same firewall instance.

---

2. Identity-Based Access & Captive Portal (NAC)

Since true 802.1X dynamic VLAN assignment can be complex to simulate inside

VirtualBox without a managed virtual switch, the industry-standard way to enforce Network Access Control (NAC) in OPNsense is via a **Captive Portal with Identity Authentication**.

Setup Steps

1. Navigate to **Services** ➡️ **Captive Portal** ➡️ **Administration**.
2. Click the **+** button to create a new zone named `HQ_NAC`.
3. Set the **Interfaces** to `HQ_LAN`.
4. Set the **Authentication Server** to `Local Database` (or a RADIUS server if you install one on Ubuntu).
5. Click **Save** and **Apply**.

📸 ***Screenshot Requirement:*** *Take a snapshot of the OPNsense Captive Portal administration screen showing the zone enabled on the HQ_LAN interface.*

📸 ***Verification Screenshot:*** *Open a browser on the Ubuntu VM. Try to go to any website, and capture the OPNsense login splash page blocking internet access until credentials are typed.*

---

3. Role-Based Access Control (RBAC) in OPNsense

You need to create two distinct administrative user tiers within the OPNsense web GUI to satisfy the RBAC requirement.

User Definition Matrix

- **Network Admin (`net_admin`):** Full read/write access to all firewall settings, VPNs, and system dashboards.
- **Security Auditor (`sec_auditor`):** Read-only access to view logs, firewall rules, and VPN status. No modification rights.

Setup Steps

1. Navigate to **System ➡️ Access ➡️ Groups**. Click **+** to add a group named `Auditors`.
2. Edit the `Auditors` group, scroll to **Assigned Privileges**, and check the boxes *only* for pages containing "Diagnostics", "Log Files", and "Status". Do not grant "All pages" or configuration rights.
3. Navigate to **System ➡️ Access ➡️ Users**.
4. Create user `net_admin` and add them to the default `admins` group.
5. Create user `sec_auditor` and add them to the `Auditors` group.

📸 ***Screenshot Requirement:*** *Capture the "Assigned Privileges" window of the `Auditors` group showing restricted page permissions.*

📸 ***Verification Screenshot:*** *Log in as `sec_auditor` from your Kali VM browser, attempt to click "Save" on a firewall rule, and screenshot the resulting access denied/read-only alert.*

---

4. Basic Site-to-Site VPN (IPsec)

To fulfill a Site-to-Site VPN using a single OPNsense firewall, you will configure a "loopback" or "cross-interface" IPsec tunnel between your simulated `hq_lan` network subnet (e.g., `192.168.10.0/24`) and your `branch_lan` network subnet (e.g., `192.168.20.0/24`).

Step-by-Step Profile

1. Navigate to **VPN ➡️ IPsec ➡️ Tunnel Settings**. Click **+** to create a Phase 1.
2. **Phase 1 Configuration (HQ Side):**
    - **Connection method:** Start immediate
    - **Key Exchange:** IKEv2
    - **Remote gateway:** IP address of your Branch WAN/Interface link
    - **Authentication:** Pre-Shared Key (Set a strong key like `OPNsenseVPNSecureKey2026!`)
    - **Encryption:** AES-256 | **Hash:** SHA256 | **DH Group:** 14 (2048 bit)
3. **Phase 2 Configuration (Data Tunnel):**
    - Click the **+** button inside your new Phase 1 entry to add Phase 2.
    - **Mode:** Tunnel IPv4
    - **Local Network:** LAN subnet (`192.168.10.0/24`)
    - **Remote Network:** Type in your Branch subnet (`192.168.20.0/24`)
    - **Protocol:** ESP
    - **Encryption:** AES-256 | **Hash:** SHA256

(Note: Create a mirrored IPsec Tunnel configuration reversing the Local and Remote networks to represent the Branch side connecting back to HQ).

📸 ***Screenshot Requirement:*** *Navigate to **VPN ➡️ IPsec ➡️ Status Overview**. Take a screenshot showing both Phase 1 and Phase 2 entries highlighted with green checkmarks/icons showing an **INSTALLED / CONNECTED** state.*

---

5. Final Documentation Outline

When you compile your final report, structure it around these clean sections:

1. **Topology Diagram:** A screenshot of your VirtualBox network configuration settings or a drawn diagram showing how `hq_lan` and `branch_lan` connect through OPNsense.

2. **Identity Verification:** Your Ubuntu browser screenshot showing the Captive Portal redirect.
3. **RBAC Validation:** Side-by-side screenshots of the OPNsense dashboard menu options when logged in as `net_admin` versus `sec_auditor`.
4. **Traffic Audit:** Open a terminal on Kali Linux, run a ping across the VPN subnet, and take a screenshot of **OPNsense Firewall Logs (Firewall ➡️ Log Files ➡️ Live View)** showing traffic successfully passing through the IPsec interface.